TrackScooter — Plate Cutting Sheet · REV 011

Rev 011 = the no-cut pod (factory rim, 11T) with the hub lowered 20 mm for more ground contact · 2026-08-10 · quantities for the WHOLE VEHICLE (2 pods) · all dimensions mm · source of truth: archive/rev011-uncut-11T-lowB/apollo_track_pod_rev011.scad

What changed vs Rev 009: B 170→150 (ground contact 203.5→231.2 mm, vehicle 20 mm lower), drop 38→22, shock angle 72→80°, shock station 47.0→51.1. **NEW: chamfer the two REAR corners of every arm bar 10×10 mm** — they are the closest thing to the spinning sprocket at full droop. Hardware (studs, shocks, springs, tubes, bearings, printed sprocket) is UNCHANGED.

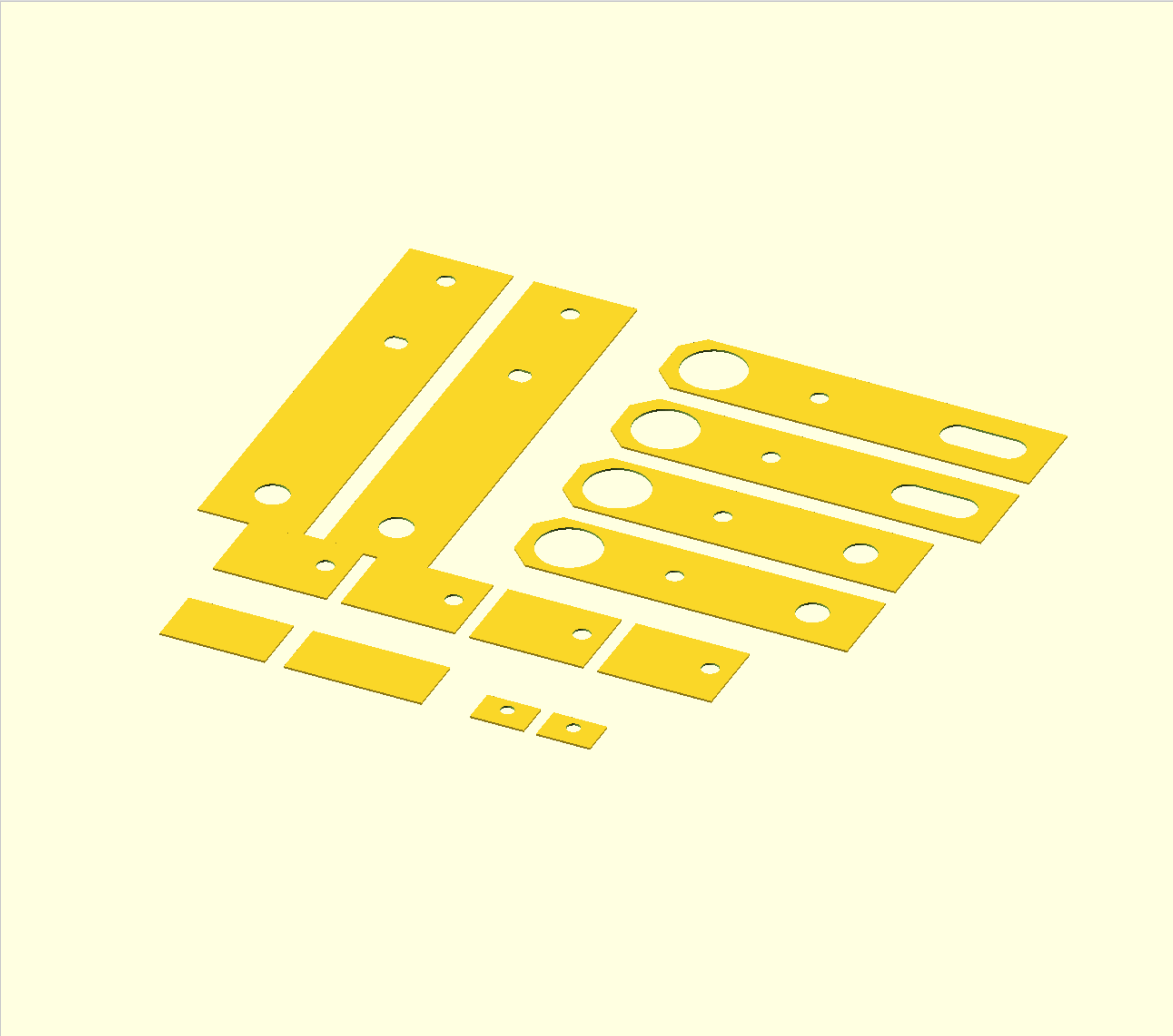
Cut list

Plate	Qty	Stock	Cut length	Holes / features — all on the bar centreline unless noted
Carrier strips	4	`50 × 6`	**`220`**	datum = hub-axle slot centre, **68 from the top end**. M8 fork-leg hole at **+52** (Ø8.5) · hub-axle slot at 0 (Ø10.4 drilled, then file two flats 8.9 apart) · pivot bore at **−128** (Ø16, ream in clamped pairs). No keel hole.
Arm bars, trailing	4	`40 × 6.35`	**`184`**	from the pivot end: pivot bore centre at **22** (Ø30.8 = Ø30 hole-saw + file to a tap-in fit of the Ø31 boss tube) · shock bolt at **73.1** (Ø8.4) · slot-end holes at **139.7** and **164.7** (Ø15.4, saw + file the 25 mm slot) · **chamfer both rear corners 10×10**
Arm bars, leading	4	`40 × 6.35`	**`166`**	pivot bore at **22** · shock bolt at **73.1** · plain axle bore at **139.7** (Ø15.4) · **chamfer both rear corners 10×10**
Cross-brace webs ("arms join")	2 + 2	`40×6 offcut`	`2×` **`50.8`** `(trailing)` `2×` **`66.5`** `(leading)`	no holes — trimmed to ~30 wide, welded between each arm's two plates spanning 22–52 from the pivot. The leading fork is wider because the forks nest at the pivot.
Tensioner lugs ("lug plate")	4	`40 × 6`	**`40 × 20`** `(one chop)`	Ø6.5 (M6 clearance) centred on the 20 mm width; welded on the trailing plates' OUTER faces with the hole 34 behind the nominal axle centre; extra length points REARWARD
Shock tab stubs	8	`40 × 6`	**`55`**	Ø8.4 pin hole; lap-welded 17 mm onto the carrier OUTER face at hub level. Position so the upper eye lands at **(±52, +2)** from the hub centre — 10 mm higher than Rev 009 because the shock now stands more upright.

Shop rules: Square ends only matter where they carry load or weld: file the tube spacers and axle sleeves square (they are clamped columns — a slanted end cocks the stack and it works loose), and the brace webs / tab stubs reasonably square for an even weld gap. The arm bars and carrier strips have FREE ends — cut, deburr, move on; what matters there is hole position, not the end face. Print `plates_rev011.dxf` 1:1, glue to the bar, centre-punch, pilot-drill, then open to size with the bars clamped in stacks (all 8 arm bars share the 22 / 73.1 / 139.7 stations) · pivot bores reamed in pairs · oil everything above

Ø10 · the chamfers are a grinder pass, do them before drilling so the bars still stack flat. **Drill bits needed:** 3–4 pilot, 6.5, 8.5, 10.5, 15.5, and a Ø30 hole saw.

Layout — ONE POD (cut everything twice for the vehicle)



The DXF now carries the COMPLETE cut list: 2 carriers · 4 arm bars (2 slotted trailing, 2 leading) · 4 shock tab stubs · 2 brace webs (50.8 trailing + 66.5 leading) · 2 tensioner lugs. Earlier revisions omitted the last two. For 1:1 paper templates with a print-scale ruler, use **rev011_1to1_drilling_templates.pdf** in this folder.